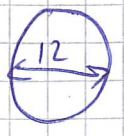
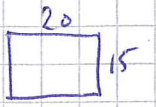
1 

$F = 18000\,N$ Sökt σ

$$\sigma = \frac{F}{A} = \frac{18000}{6^2 \cdot \pi} = 159\ N/mm^2$$

2. 

$$\sigma = \frac{40000}{15 \times 20} = 133\ N/mm^2$$

3 

$$\sigma = \frac{4000}{5 \cdot 5} = 160\ N/mm^2$$

4 

$$\sigma = \frac{10000}{10^2 \cdot \pi} = 32\ N/mm^2$$

5. 

$\sigma = \frac{F}{A} \Rightarrow F = \sigma \cdot A \Rightarrow 2 \cdot (70 \cdot 30) = \underline{4200\,N}$

6. $F = 10000$ $\sigma_{max} = 20\ N/mm^2$ sökt d:

$\sigma = \frac{F}{A} \Rightarrow A = \frac{F}{\sigma} = \frac{10000}{20} = 500$ $A = \pi r^2 \Rightarrow r = \sqrt{A/\pi} \Rightarrow$

$\Rightarrow r = \sqrt{500/\pi} = 12{,}6 \Rightarrow d = 2r = 25\,mm$ Avrunda uppåt!

7. $d = 10\,mm$ ○ $\sigma_{max} = 80\ N/mm^2$ Sökt: F

$\sigma = \frac{F}{A} \Rightarrow F = \sigma A \Rightarrow 80 \cdot 5^2 \pi = 6283\,N$

8. $\varepsilon = \frac{\Delta L}{L_0} = \frac{20}{4000} = 0{,}005 \Rightarrow 0{,}5\%$

9. $\varepsilon = \frac{\Delta L}{L_0} \Rightarrow \Delta L = \varepsilon \cdot L_0 = 0{,}0005 \cdot 3000 = 1{,}5\,mm$

$L_{tot} = L_0 + \Delta L = 3001{,}5\,mm$

10. (12) $L_0 = 5128\,mm$ $d = 12\,mm$ $F = 9000\,N$ $L_{tot} = 5130\,mm$

a) $\Delta L = 2\,mm$

b) $\varepsilon = \frac{\Delta L}{L_0} = \frac{2}{5128} = 3{,}9 \cdot 10^{-4}$

c) $\sigma = \frac{F}{A} = \frac{9000}{6^2 \cdot \pi} = 79{,}6\,N/mm^2$

11. $\varepsilon = 0{,}27$ $L_0 = 125\,mm$ sökt: L_{tot}

$\varepsilon = \frac{\Delta L}{L_0} \Rightarrow \Delta L = \varepsilon \cdot L_0 = 0{,}27 \cdot 125 = 33{,}75$ $L_{tot} = 33{,}75 + 125 = 158{,}75$

12.  $\sigma = \frac{F}{A} = \frac{12}{1 \cdot 4} = 3\,N/mm^2$

13 $L_0 = 70\,mm$ $\Delta L = 0{,}1\,mm$

$\varepsilon = \frac{0{,}1}{70} = 0{,}0014$

14. $L_{tot} = 127\,mm$ $\varepsilon = 0{,}29$ sökt L_0.

$\varepsilon = \frac{\Delta L}{L_0}$ $L_{tot} \% = \varepsilon + 1{,}00 = 1{,}29 \Rightarrow L_0 \frac{127}{1{,}29} = 98{,}5\,mm$

15. $R_{e1} = 260\ N/mm^2$

$R_m = 500\ N/mm^2$

$\sigma = 65\ N/mm^2$

a) $\frac{260}{65} = 4$ 4ször

b) $\frac{500}{65} = 7{,}7$ 5ször

16. $F = 5000N$

$n_s = 3$ kerekítsünk

$R_e = 160$

$\sigma_{hú} = \frac{R_e}{n_s} = \frac{160}{3} = 53$

$53 = \frac{5000}{A} \Rightarrow A = \frac{5000}{53} = 93{,}75\ mm^2$

$A = s \cdot s \Rightarrow s = \sqrt{A} = 9{,}7mm$

17. 

$A = \pi r^2 \Rightarrow r = \sqrt{\frac{93{,}75}{\pi}} = 5{,}46 \Rightarrow d = 11mm$

18. $F = 10000$ $\sigma_{hú} = 5\ N/mm^2$

$A = ?$

a) $\sigma_{hú} = \frac{F}{A} \Rightarrow A = \frac{F}{\sigma_{hú}} = \frac{10000}{5} = 2000\ mm^2$

b) $2000 = x \cdot 3x \Rightarrow x = \sqrt{\frac{2000}{3}} = 26mm$

$Dim = 26 \times 78$

19. $F = 40000$ Lina

$R_m = 1200\ N/mm^2$ $n_s = 8$

Sökt: d

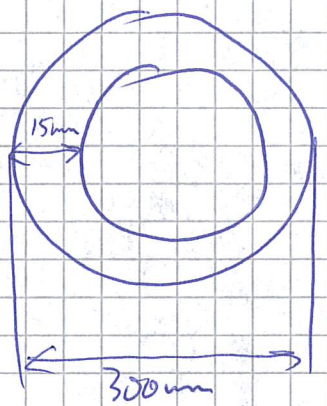
$\sigma_{till} = \frac{1200}{8} = 150$

F/tråd $40000/100 = 400$ skall vara $\leq$ än upp

$\sigma_{till} = \frac{F}{A} \Rightarrow A = \frac{F}{\sigma_{til}} = \frac{400}{150} = 2{,}67\ mm^2$

$d = \sqrt{\frac{2{,}67}{\pi}} \cdot 2 = 1{,}84\ mm$

20.



$F = 800000$

$A = 150^2 \cdot \pi - 135^2 \pi = 13430\ mm^2$

$\sigma = 800000/13430 = 59{,}6\ N/mm^2$

21. $F = 1{,}2 \cdot 10^4$ $n_s = 4$ $R_{p0,2} = 160\ N/mm^2$ $\square$ $s = ?$

$\sigma_{till} = R_{p0,2} = \frac{160}{4} = 40\ N/mm^2$ $40 = \frac{12000}{5^2} \Rightarrow$

$s = \sqrt{\frac{12000}{40}} = 17{,}4\ mm$

22. $d=5\,mm$ $F=1500\,N$ $L=20\,m$ $E_{Cu}=1{,}1\cdot10^5$

$\sigma=E\cdot\varepsilon$ $\sigma=\frac{F}{A}$

$\frac{F}{A}=E\cdot\varepsilon$ $\frac{F}{2{,}5^2\pi}=1{,}1\cdot10^5\cdot\frac{\Delta L}{L_0}$

$\frac{1500}{2{,}5^2\pi}=1{,}1\cdot10^5\cdot\frac{\Delta L}{20000}$

$\Delta L=\left(\left(\frac{1500}{2{,}5^2\pi}\right)/1{,}1\cdot10^5\right)\cdot 20000=\underline{\underline{13{,}9\,mm}}$

---

23 Sökt: E



$\Delta L=0{,}6\,mm$ $\sigma=120\,N/mm^2$

$E=\frac{\sigma}{\varepsilon}\Rightarrow\frac{120}{6\cdot10^{-4}}=200000\,N/mm^2$

$\varepsilon=\frac{\Delta L}{L_0}=\frac{0{,}6}{1000}=6\cdot10^{-4}$

---

24 $L_0=2000\,mm$ $d=1{,}5\,mm$ $\Delta L=4{,}5\,mm$ $F=490$

{ $R_{e1}=500\,N/mm^2$ } Sökt: E

$E=\frac{\sigma}{\varepsilon}$ ~~$\sigma=\frac{F}{A}$~~ $\varepsilon=\frac{\Delta L}{L_0}$ $\varepsilon=\frac{4{,}5}{2000}=0{,}0023$

$E=\frac{R_e(F/A)}{(\Delta L/L_0)}=\frac{490\cdot2000}{0{,}75^2\cdot\pi\cdot4{,}5}=1{,}23\cdot10^5\,N/mm^2$ (500 / $2{,}2\cdot10^5$)

25 $d = 22\,mm$ $1/20\% = 0{,}05\% = 0{,}0005 = \Delta L$

$F = 40000\,N$ sökt: E

$$\varepsilon = \frac{\Delta L}{L_0} \qquad \sigma = \frac{F}{A} \qquad \sigma = E \cdot \varepsilon$$

$$\text{sätt: } \frac{F}{A} = E \cdot \varepsilon \Rightarrow E = \frac{F}{A \cdot \varepsilon} \Rightarrow$$

$$E = \frac{40000}{11^2 \cdot \pi \cdot 0{,}0005} = 2{,}1 \cdot 10^5\ N/mm^2$$

---

26 $L_0 = 3500\,mm$ $d = 2{,}5\,mm$ $F = 800\,N$ $\Delta L = 2{,}8\,mm$

sökt: E

$$E = \frac{800}{1{,}25^2 \cdot \pi \cdot \left(\frac{2{,}8}{3500}\right)} = 2{,}04 \cdot 10^5\ N/mm^2$$

---

27 $A = B \cdot L$ $B = 20\,mm$ $F = 8000$ $\sigma = 9\ N/mm^2$

~~$\sigma_{till} = \frac{\sigma_{brott}}{n}$~~ ~~$\sigma = 0{,}6\,\sigma_{brott}$~~ $\Rightarrow \sigma_{brott} = \frac{9}{0{,}6} = 15\ N/mm^2$

$9 = \frac{8000}{20 \cdot L} \Rightarrow L = 444$

$L = \frac{8000}{9 \cdot 20} = 44{,}4$

---

28 $L = 45\,mm$ $B = 30\,mm$ $\tau_{till} = 13\ N/mm^2$

~~$\tau_{till} = 0{,}6\,\sigma$~~ $\Rightarrow 13 = 0{,}6\,\sigma$ $\sigma = \frac{13}{0{,}6} = 21{,}67$

sökt: F

$$F = \tau \cdot A = 13 \cdot 45 \cdot 30 = 17550\ N$$

29 $d = 30$ mm 2 fogar $\tau = 13$ N/mm² sökt: F

$\tau_{hm} = \frac{F}{A}$

$\tau_{hm} = 0{,}6\,\sigma_{fk}$ $\sigma_{fk} = \frac{13}{0{,}6} = 21{,}67$ N/mm²

$F = \sigma_{fk} \cdot A = 21{,}67 \cdot 2 \cdot 15^2 \cdot \pi = 30630$ N 18378 N

---

30 $F = 18000$ N SS 1325 $R_e = 320$ N/mm² $n_s = 1{,}6$

Sökt: d välj sedan på s. 29

$\sigma_{fk} = \frac{R_e}{n_s} = \frac{320}{1{,}6} = 200$ N/mm² $\Rightarrow \tau_{hm} = 120$ N/mm²

$\tau_{hm} = \frac{F}{A} \Rightarrow A = \frac{F}{\tau_{hm}} = \frac{18000}{120} = 150$

4 muttrar = 150 mm² 1 mutt = 37,5 mm²

$A = \pi r^2 \Rightarrow 37{,}5 = \pi \cdot r^2 \Rightarrow r = \sqrt{\frac{37{,}5}{\pi}} = 3{,}45 \Rightarrow d = 6{,}9$ mm

$\Rightarrow$ mutt 8 mm

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31 $F = 25000$ $n_s = 3$ $R_e = 280$ N/mm² sökt d

$\tau_{hm} = 280 \cdot 0{,}6 = 168$ N/mm² $\sigma_{fk} = \frac{168}{3} = 56$ N/mm²

$\tau_{hm} = \frac{F}{A_1} \Rightarrow A_1 = \frac{25000}{56} = 446$ mm²

1 mutt $\frac{A_s}{2} = 223$ mm² $r = \sqrt{\frac{223}{\pi}} = 8{,}42 \Rightarrow d = 16{,}8 \Rightarrow$ 19 mm (avrundar)

32. $d = 1{,}5\,mm$ $\tau_m = 350\ N/mm^2$

~~$\tau_m = 0{,}6 \cdot 350 = 210\ N/mm^2$~~

$F = \tau \cdot A_s \Rightarrow 350 \cdot 0{,}75^2 \pi = 618\,N \approx 620$

33. $\tau_m = 250\ N/mm^2$ (on books) $d = 25\,mm$ $t = 10\,mm$ $n_b = 4$

$A_s = 25 \cdot \pi \cdot 10$

$\tau_m = \frac{F}{A_s} \Rightarrow F = \tau \cdot A_s = 250 \cdot 25\pi \cdot 10 = 196\,kN$

R_b $\sigma_b = \frac{F}{A} = \frac{R_b}{n_b} \Rightarrow R_b = \frac{F \cdot n_b}{A} = \frac{196000 \cdot 4}{12{,}5^2 \pi} = 1597\ N/mm^2$

34. $t = 10\,mm$ $\tau_n = 350\ N/mm^2$ $d = 30\,mm$

$A_s = 30 \cdot \pi \cdot 10 = 942\ mm^2$

$\tau = \frac{F}{A} \Rightarrow F = 350 \cdot 942 = 329867\,N$

$\sigma = \frac{F}{A} \Rightarrow \frac{329867}{15^2 \pi} = 467\ N/mm^2$

35 $F = ?$ $d = 35\,mm$ $t = 3\,mm$ ~~$R_e = 350$~~ $R_e = 380\ N/mm^2$

$\tau_s = 380 \cdot 0{,}6 = 228$ $\tau = \frac{F}{A_s} \Rightarrow F = 228 \cdot 35 \cdot \pi \cdot 3 =$

$= 75\,kN$

36. $F = 300000$ Skiål $d = 20\,mm$ $R_B = 440\ N/mm^2$

$\tau = 440 \cdot 0{,}6 = 264$

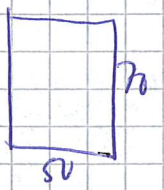
Sökt t $\quad \tau = \frac{F}{A} \Rightarrow \tau = \frac{300000}{A} \Rightarrow$

$A = \frac{300000}{264} = 1136 \qquad A = 5 \cdot 20 \cdot \pi \cdot t \Rightarrow$

$t = \frac{1136}{5 \cdot 20 \cdot \pi} = 3{,}61\ mm$

37 4 | hösthside

42



$W_b = \frac{BH^2}{6} = \frac{50 \cdot 70^2}{6} = 40833\ mm^3$

$\frac{70 \cdot 50^2}{6} = 29167\ mm^3$

43 $B = 35\,mm$ Sökt H $\quad W_b = 36000$

$36000 = \frac{35 \cdot H^2}{6} \Rightarrow \sqrt{\frac{36000 \cdot 6}{35}} = H = 78{,}6\ mm$

44. (63) $\quad W_b = \frac{\pi D^3}{32} = \frac{\pi \cdot 63^3}{32} = 24550\ mm^3$

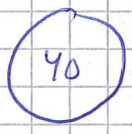
45 $W_b = 24000 \Rightarrow d = \sqrt[3]{\frac{W_b \cdot 32}{\pi}} = 62{,}5\,mm = 63\,mm$

46



$W_b = \frac{\pi(65^4 - 57^4)}{32 \cdot 65} = 11017\ mm^3$

47



Sökt d $\quad W_b = 3700\ mm^3$

$3700 = \frac{\pi(40^4 - d^4)}{32 \cdot 40} \Rightarrow \frac{3700 \cdot 32 \cdot 40}{\pi} = 40^4 - d^4 \Rightarrow$

$\sqrt[4]{40^4 - \frac{3700 \cdot 32 \cdot 40}{\pi}} = d = 32\,mm$